



Technical Report

Life Expectancy Analysis and Prediction Using Modeling and Simulation

Course Title: Introduction to Modeling and Simulation

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1 Introduction

Life expectancy is one of the important indicators of the development in healthcare and quality living standards of a country. And so analyzing historical life expectancy data can help us identify long-term trends, compare countries, and even make informed predictions about future outcomes.

The essence of this assignment was to let us use Python to retrieve life expectancy data, visualize trends, clean the dataset, compute descriptive statistics, and develop predictive models. For the assignment, we compared Ghana, Mauritius, and Nigeria as directed from the assignment prompts.

We went on to develop two predictive models (Linear Regression and Polynomial Regression) to forecast the life expectancy up to the year 2050 for Ghana. This assignment allowed us to practice how modeling and simulation techniques can be applied to real-world public health data.

2 Dataset Description

We used the dataset that was provided in the assignment prompts. The dataset has life expectancy records for multiple countries over several decades.

Each record in the dataset contains:

- Entity (Country name)
- Code
- Year
- Life expectancy (years)

The dataset spans many decades and includes data for Ghana up to 2023. For comparison purposes, we ensured that Ghana and Mauritius were compared only across years common to both countries. Nigeria was added as a third comparison country.

Generally, the data shows gradual improvement in life expectancy over time. It shows global improvements in healthcare, nutrition, sanitation, and disease control.

3 Methodology

This section explains how the analysis and modeling were done.

3.1 a) Data Retrieval and Visualization

Using pandas:

- We filtered data for Ghana and Mauritius.
- Merged both datasets on common years.
- Plotted both countries on the same graph for comparison.
- Added Nigeria as a third country and plotted all three together.

The graphs were labeled clearly and decorated as required in the assignment prompt.

3.2 b) Data Cleaning

Missing Values

We checked for missing values using:

```
isna().sum()
```

Very few missing values were found.

Outliers / Suspicious Jumps

We defined suspicious jumps as any change greater than 5 years between consecutive years. We calculated year-to-year differences and flagged large jumps.

Ghana had a jump of 15.07 years in 1950. Mauritius had a jump of 13.32 years in 1950 (compared to 1947).

No other major unrealistic jumps were observed.

Handling Missing Data

We used linear interpolation. The reasons are that:

- Life expectancy changes gradually over time.
- Interpolation maintains realistic trends.
- It avoids artificially dropping valuable data.

3.3 c) Before vs After Cleaning

We plotted graphs before and after cleaning. The graphs were nearly identical, confirming that the dataset was already stable and reliable.

3.4 d) Statistical Analysis

For each country, we computed:

- Mean
- Median
- Standard Deviation
- Minimum
- Maximum
- Growth rate per decade

We grouped years into decades and calculated the percentage change in life expectancy from one decade to the next.

3.5 e) Predictive Modeling (Ghana Only)

We selected two models:

Linear Regression

Assumes a constant rate of increase over time.

Polynomial Regression (Degree 3)

Allows curved growth patterns and captures non-linear trends.

Steps performed:

- Trained both models using historical Ghana data.
- Predicted life expectancy up to 2050.
- Plotted both predictions on the same graph.
- Compared model behavior.

4 Results

4.1 Country Comparison

- Mauritius consistently had the highest life expectancy.
- Ghana performed better than Nigeria.

- All three countries showed long-term improvement.

Mauritius started around 52 years in 1950 and reached nearly 75 years by 2023. Ghana started around 43 years and reached about 65 years by 2023. Nigeria started around 36 years and reached roughly 54 years by 2023.

Mauritius's higher performance suggests stronger healthcare systems and socio-economic conditions.

4.2 Data Cleaning Results

- Missing values were minimal.
- No extreme suspicious jumps were found apart from the early 1950 records.
- Interpolation slightly smoothed the data.
- Before and after cleaning graphs were almost identical.

This confirms that the dataset was of good quality.

4.3 Statistical Findings

Mean and Median

- Mauritius had the highest average life expectancy.
- Nigeria had the lowest.
- Ghana was in between.
- The closeness between mean and median suggests relatively stable distributions.

Standard Deviation

Moderate variation indicates steady improvements rather than erratic changes.

Growth Rate per Decade

Growth rates were higher in earlier decades, especially during periods of improved vaccination and disease control.

Note: The first decade for each country shows NaN growth rate because there is no previous decade to compare against.

4.4 Predictive Modeling Results

Linear Regression

- Shows steady, constant growth.
- Simple and easy to interpret.
- May underestimate or overestimate if growth is not perfectly linear.

Polynomial Regression

- Captures curvature in the trend.
- Fits historical data more closely.
- Provides a more realistic long-term projection.

Both models predict continued increase in Ghana's life expectancy up to 2050.

5 Discussion

The results reveal several important insights.

First, Mauritius clearly outperforms both Ghana and Nigeria in life expectancy. This likely reflects differences in healthcare infrastructure, economic stability, and public health investments.

Second, Ghana's performance is stronger than Nigeria's, suggesting relatively better improvements in health outcomes over time.

Third, the dataset was largely clean, which increases confidence in our analysis.

When comparing predictive models:

- The linear model assumes constant growth.
- The polynomial model adapts better to non-linear changes.

Because life expectancy does not increase in perfectly straight lines, the polynomial model provides a more flexible and realistic prediction. Therefore, we consider the polynomial regression model to be the better model for long-term forecasting.

However, predictions assume current trends continue. Unexpected events such as pandemics, economic crises, or major healthcare reforms could significantly change actual outcomes.

6 Limitations

Despite the strong analysis, several limitations exist:

1. The models use only Year as a predictor.
2. They do not account for GDP, healthcare spending, education, or disease outbreaks.
3. Polynomial regression may overfit historical data.
4. Long-term predictions (to 2050) carry uncertainty.
5. Growth rate calculations show NaN in the first decade because percentage change requires a previous value.
6. Real-world factors like COVID-19 or future policy shifts are not included in the model.

7 Conclusion

This project successfully applied modeling and simulation techniques to real-world life expectancy data.

Key conclusions:

- Mauritius consistently has the highest life expectancy among the three countries.
- Ghana performs better than Nigeria but still lags behind Mauritius.
- All countries show steady long-term health improvement.
- The dataset was clean and reliable.
- Polynomial regression provided a better fit for predicting Ghana's life expectancy to 2050.
- Ghana's life expectancy is expected to continue increasing if current trends persist.

Overall, this assignment demonstrated how Python, statistical analysis, and predictive modeling can be used to analyze public health data and make informed projections about the future.